Supplementary Information:

Racing amidst change: urbanization and climate alter functional traits and distribution of an Amazonian parthenogenetic lizard

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1. Materials and Methods

1.1. Study area & specimens sampling

The survey was realized under SISBIO licenses n. 83159-1/2022 and nº 44832-4. The physiological experiments were realized under CEUA/INPA license n. 042/2020 SEI 01280.0001606/2020-88.

The Amazon Forest is often seen as an ocean of trees with wide rivers crossing it, however, it is much heterogeneous, composed by different phytophysiognomies, such as open areas ("Campinas", Savannas), "Campinaranas", "Terra Firme" and "Igapó" (flooded area) (Braga, 1979; Adeney et al., 2016; de Carvalho & Mustin, 2017). The campinas and campinaranas are regions with isolated small bushes and arboreal species, respectively (Adeney et al., 2016; Guimarães & Bueno, 2016). Their soil presents mostly sand content with a small clay component, and has low drainage capacity, becoming seasonally flooded in some localities (Adeney et al., 2016; Guimarães & Bueno, 2016). In northern Amazonia, the savannas of Roraima are an enclave of open grasslands ("campo limpo", "campo sujo") with a mosaic of shrubs, scattered trees, and forest patches (Barbosa & Miranda, 2004). Thus, we collected individuals and ecological and physiological data in two localities in Brazil with contrasting environments (see Figure S1): Manaus – Amazonas, (hereafter referred to as the neonative population; see Essl et al., 2019), on 03-27, August 2022, and 16-18 January 2023 during the rainy and dry season; and Bonfim – Roraima (hereafter referred to as native population), on 17-22 July 2022, during the rainy season. The precipitation differs between dry and rainy seasons, with accumulated precipitation of 56.1 mm to 347 mm respectively (INMET, 2022). These regions differ in temperature and precipitation. Manaus has a mean of 27.3°C (minimums of 23.7°C and maximum of 34.1°C) and annual precipitation of 2,300 mm. In contrast, Roraima's savannas have a mean of 27.8°C (maximum of 34.3°C and minimums of 22.4°C) and annual precipitation of 1,600 mm (Braga, 1979; Barbosa & Miranda 2004; Corrêa et al., 2016; INMET, 2022). However, the urban region of Manaus unveiled an increase in temperature of 0.7°C from 1961 to 2008 (Souza & Alvalá, 2014) and might be even hotter nowadays. The soil of Manaus and savannas of Roraima are mainly composed of clay, with some addition of sand and silt (Chauvel et al., 1987; IBGE, 2025).

The morphological identification of individuals was made in the field following Recoder et al. (2018) and confirmed with molecular analysis using mtDNA 16S (see below). Lizards were caught through active surveys during the hottest hours of the day (1000 h to 1400 h) in open areas in Bonfim, and next to buildings or/and garden areas in Manaus (Cole et al., 1989; Vitt & Zani, 1998), as these lizards were observed active in the hottest hours of the day even on hot surfaces (Cole et al., 1990). To enhance the sampling effort in Manaus, since *'[…] on a good day of collecting, four field workers might catch a total of eight or ten of these lizards […]'* (Cole et al., 1989), we used five miniaturized pitfall traps with about four 1-liter buckets between a fence of 4 m × 30 cm (Figure 1).

Figure 1. Pitfall traps used to collect individuals of *Gymnophthalmus underwoodi* Grant 1958, in Manaus-AM, Brazil. Photos and drawings by the first author.

1.2. Genetic data

To confirm the identification of sampled individuals with greater accuracy, we extracted genetic samples of the mitochondrial marker 16S (± 550 bp) from tissues of the tail muscle and liver of eight collected specimens from Amazonas and Roraima (Table 1). The DNA was extracted following Promega Wizard® Genomic DNA purification Kit. The16S was amplified following Pellegrino et al., (2001) using primers 16SF (5'-CTGTTTACCAAAAACATMRCCTYTAGC-3') and 16SR (5'-TAGATAGAAACCGACCTGGATT-3'). The purification followed PEG 8000 protocol (Sambrook & Russel, 2001) and was submitted to sequencing following BigDye Terminator 3.0 Cycle Sequencing kit (Applied Biosystems). Additionally, we used all sequences of the species *Gymnophthalmus underwoodi* available in Genbank for phylogenetic analysis, and sequences of *Micrablepharus atticolus* Rodrigues 1996, *M. maximiliani* (Reinhardt & Lütken, 1862) and *Calyptommatus sinebrachiatus* Rodrigues 1991, as outgroups (Table 1) aligning with Geneious v7.0. To infer the phylogenetic tree, we used MrBayes v.3.2 (Ronquist et al., 2012) through the CIPRES webserver (http://www.phylo.org/). The GTR+G was chosen as the best substitution model for the alignment, obtained using MrModelTest v2.4 (Nylander et al., 2008). The Bayesian Inference (BI) was made through two independent runs of 50 million generations each, four Markov chains, and trees sampling every 5,000 generations (Recoder et al., 2018). The *burn-in* of 10% was applied using Tracer v1.7.2 and the final tree was fitted with FigTree v1.4.4.

81 Table 1. Molecular sampling and sequences retrieved from Genbank. Specimens used in this study are in process of deposit in Genbank.

Voucher	**Species**	**Locality**	**Country**	**16S**	**Reference**
AMNH-R-138374	*Gymnophthalmus cryptus*	San Juan de Manapiare, Amazonas	Venezuela	AF101362	Kirizian & Cole, 1999
MTR 946613	*Gymnophthalmus leucomystax*	Alto Alegre, Roraima	Brazil	AF420715	Pellegrino et al., 2001
AMNH-R-139856	*Gymnophthalmus leucomystax*	Rupununi Savanna	Guyana	AF101363	Kirizian & Cole, 1999
AMNH-R-139857	*Gymnophthalmus leucomystax*	Rupununi Savanna	Guyana	MH732714	Recoder et al., 2018
MTR 946618	*Gymnophthalmus leucomystax*	Alto Alegre, Roraima	Brazil	MH732715	Recoder et al., 2018
AMNH-R-128428	*Gymnophthalmus pleii*	Martinique	West Indies	AF101364	Kirizian & Cole, 1999
MTR 33465	*Gymnophthalmus speciosus*	San José	Costa Rica	MH732710	Recoder et al., 2018
ALM 5979	*Gymnophthalmus* cf. *speciosus*	Adicora, Falcón	Venezuela	AF101367	Kirizian & Cole, 1999
JC/FT7054	*Gymnophthalmus underwoodi*	St. Phillip	Barbados	AF101369	Kirizian & Cole, 1999
APL 21703	*Gymnophthalmus underwoodi*	Manaus, Amazonas	Brazil	MH732712	Recoder et al., 2018
INPA-H044762	*Gymnophthalmus underwoodi*	Manaus, Amazonas	Brazil	OR438308	This study
INPA-H044761	*Gymnophthalmus underwoodi*	Manaus, Amazonas	Brazil	OR438309	This study
INPA-H044753	*Gymnophthalmus underwoodi*	Manaus, Amazonas	Brazil	OR438310	This study
INPA-H044769	*Gymnophthalmus underwoodi*	Manaus, Amazonas	Brazil	OR438311	This study
INPA-H044759	*Gymnophthalmus underwoodi*	Manaus, Amazonas	Brazil	OR438312	This study
INPA-H044760	*Gymnophthalmus underwoodi*	Manaus, Amazonas	Brazil	OR438313	This study
INPA-H044764	*Gymnophthalmus underwoodi*	Manaus, Amazonas	Brazil	OR438314	This study

INPA-H044752	*Gymnophthalmus underwoodi*	Manaus, Amazonas	Brazil	OR438315	This study
INPA-H044745	*Gymnophthalmus underwoodi*	São João da Baliza, Roraima	Brazil	OR438316	This study
INPA-H044744	*Gymnophthalmus underwoodi*	São João da Baliza, Roraima	Brazil	OR438317	This study
INPA-H044730	*Gymnophthalmus underwoodi*	Bonfim, Roraima	Brazil	OR438318	This study
INPA-H044728	*Gymnophthalmus underwoodi*	Bonfim, Roraima	Brazil	OR438319	This study
INPA-H044726	*Gymnophthalmus underwoodi*	Bonfim, Roraima	Brazil	OR438320	This study
INPA-H044732	*Gymnophthalmus underwoodi*	Bonfim, Roraima	Brazil	OR438321	This study
INPA-H044729	*Gymnophthalmus underwoodi*	Bonfim, Roraima	Brazil	OR438322	This study
INPA-H044740	*Gymnophthalmus underwoodi*	Bonfim, Roraima	Brazil	OR438323	This study
INPA-H044734	*Gymnophthalmus underwoodi*	Bonfim, Roraima	Brazil	OR438324	This study
INPA-H044735	*Gymnophthalmus underwoodi*	Bonfim, Roraima	Brazil	OR438325	This study
MTR 946590	*Gymnophthalmus underwoodi*	Ilha de Maracá, Roraima	Brazil	KT254406	Goicoechea et al., 2016
NYSM 6432	*Gymnophthalmus underwoodi*	Bottomless Ghaut	Montserrat	KX866265	Snyder et al., 2017
MPEG 33517	*Gymnophthalmus underwoodi*	Belém, Pará	Brazil	MZ544022	Maciel et al., 2021
MTR 946601	*Gymnophthalmus underwoodi*	Ilha de Maracá, Roraima	Brazil	MH732711	Recoder et al., 2018
MTR 946639	*Gymnophthalmus vanzoi*	Alto Alegre, Roraima	Brazil	AF420743	Pellegrino et al., 2001
MTR 946484	*Gymnophthalmus vanzoi*	Boa Vista, Roraima	Brazil	MH732699	Recoder et al., 2018
MTR 946500	*Gymnophthalmus vanzoi*	Normandia, Roraima	Brazil	MH732705	Recoder et al., 2018
MTR 946529	*Gymnophthalmus vanzoi*	Uiramutã, Roraima	Brazil	MH732708	Recoder et al., 2018

MPEG 29841	*Gymnophthalmus vanzoi*	Mazagão, Amapá	Brazil	MZ544020	Maciel et al., 2021
MTR 33447	*Gymnophthalmus vanzoi*	Santarém, Pará	Brazil	MH732695	Recoder et al., 2018
AMNH-R-140975	*Gymnophthalmus* aff. *vanzoi*	Berbice River	Guyana	AF101366	Kirizian & Cole 1999
AMNH-R-128438	*Gymnophthalmus* aff. *vanzoi*	St. George	Trinidad and Tobago	MH732709	Recoder et al., 2018
LG 854	*Micrablepharus atticolus**	Santa Rita do Araguaia, Goiás	Brazil	AF420718	Pellegrino et al., 2001
LG 1017	*Micrablepharus maximiliani**	Barra do Garças, Mato Grosso	Brazil	AF420730	Pellegrino et al., 2001
MRT 05054	*Calyptommatus sinebrachiatus**	Gentio do Ouro, Bahia	Brazil	AF420720	Pellegrino et al., 2001

* species set as outgroup in phylogenetic analysis

1.3. Preferential and voluntary temperatures

The collected specimens were kept individually isolated in plastic recipients with moistened paper and without access to food. These conditions were maintained for approximately four days in an isolated room with air temperature ranging from 26 to 30°C. The abstention from food was made to avoid the effects of diet on preferred temperature (T_{pref}) (Gilbert & Miles, 2016; Simandle et al., 2001; Abayarathna & Webb, 2021). The thermal tests were conducted sequentially, starting with the assessment of preferred temperature (T_{pref}), followed by sprint performance, and concluding with the determination of critical temperatures. Each individual passed through all experiments.

We determined the T_{pref} using a thermal gradient made of plastic roof lining measuring 1,00 m × 1,00 m × 0,33 m, split into eight channels with a width of 12.50 cm each (Paranjpe et al., 2012) (Fig. 2). We used a zinc plate as the base of the gradient, with ice bags at one end and 60W incandescent light bulbs covered with yellow cellophane paper at the other end to reduce the effects of color during the thermoregulation (Fleishman et al., 2011; Pérez I de Lanuza et al., 2018). The thermal gradient ranged from 40°C to 10°C in each channel. The temperature range used in this experiment aligns with those commonly applied in similar studies (e.g., Camacho et al., 2014; Diele-Viegas et al., 2018; Martins et al., 2021) and closely matches the thermal conditions experienced by the species in its natural habitat. In Manaus, air temperatures range from a minimum of 23.7°C to a maximum of 34.1°C, while in Boa Vista, they range from 22.4°C to 24.3°C. Additionally, Cole et al. (1990) observed that *G. underwoodi* individuals select areas with maximum air temperatures around 40.0°C. The individuals were acclimated for 30 minutes inside the thermal gradient before starting the measurements. We placed the lizards in the middle of the lane to reduce the chance of directional bias. Using a small thermocouple

(Omega® 5SC-TT-T-36-72 *Ready-Made Insulated Thermocouple with* Kapton®, PFA, *Glass Braid Insulation and Molded Connector*) attached to the lizards' belly with hospital tape, we recorded the lizards' body temperature every minute for 120 minutes on the gradient with a data-logger (Omega ® TC-08-TC-DAQ 8-*Channel USB Thermocouple Data Acquisition Module*). To minimize stress, as the core-surface temperature does not change significantly in small animals, the thermocouple was attached to the lizards' belly with hospital tape (Camacho et al., 2014; Diele-Viegas et al., 2018). The T_{pref} was determined as the mean of the body temperatures recorded during the experiment, with the 5th percentile representing the voluntary minimum temperature (VT_{min}), and the 95th percentile representing the voluntary maximum temperature (VT_{max}) (Caetano et al., 2020).

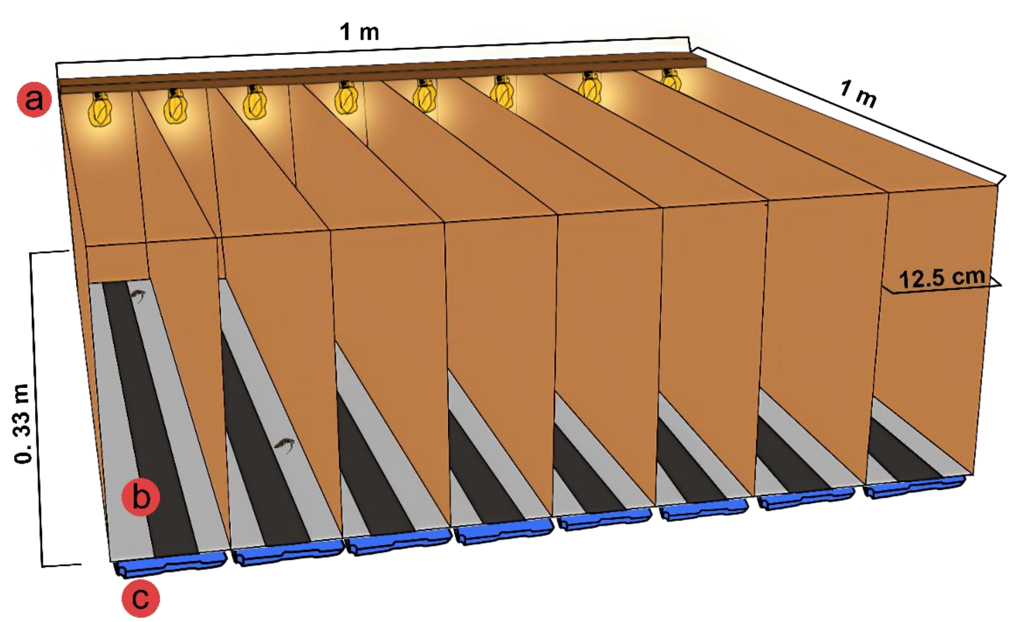

Figure 2. Structure to perform the Preferential Temperature test (T_{pref}). Support with 60W incandescent light bulbs covered with yellow cellophane paper (a); Zinc plate with sandpaper to facilitate the mobility of individuals (b); Ice bags (c). Drawings by the first author.

1.4. Sprint performance

After inferring the T_{pref}, we measured the sprint performance. We built a linear track with plastic roof lining measuring 1.00 m × 0.30 m × 0.30 m (Fig. 3). To measure the performance across different substrates, the base of the track was made with two types of substrates (herein treatments): one with red clay to represent native environment (native substrate); and the other with coarse-grain sandpaper (60 grit; grains ~400 μm) to simulate the street substrate in the urban environment (urban substrate). We selected these substrates since *G. underwoodi* is a lizard commonly found in gardens and near houses in urban environments, frequently observed active during the hottest hours of the day on warm surfaces. They are usually encountered running through the grass in gardens, but individuals can also be found on human-made surfaces such as concrete, walls, and streets. The first author recorded one individual near a street at the Universidade Federal do Amazonas (UFAM), published on iNaturalist (https://www.inaturalist.org/observations/129431075), while another observation was made between university buildings where students attend classes (https://www.inaturalist.org/observations/136613349). Another collection site was an apartment complex (-03.091350, -59.989627), where individuals had no way to enter or leave the area without crossing streets and other cemented surfaces. Additionally, we filmed an individual climbing a cement wall in an urban environment, which can be viewed at the link https://youtu.be/i4X32hPchQ0.

Other researchers have also documented individuals on human-made structures, such as streets (which we used as one of our substrates). Maciel et al. (2021) found individuals in fluvial port areas, while Alfonso et al. (2012) observed individuals basking in leaf litter between buildings on the University of Oriente campus. In their natural Amazonian habitat, these lizards inhabit leaf litter or grass in open vegetation, open forests, and sandy riverbanks (Ávila-Pires 1995; Ribeiro-Júnior & Amaral 2017). However, in the savanna region where some specimens were collected, the terrain consists mainly of open areas with predominantly clay-rich soil (https://www.ibge.gov.br/geociencias/informacoes-ambientais/pedologia/15829-solos.html; Adeney et al. 2016).

Therefore, we chose to use coarse-grain sandpaper as a substrate because urban individuals frequently traverse human-made surfaces, particularly streets and cemented areas. At the same time, we selected clay as a substrate due to its prevalence in the savanna region and the limitations of using grass in experimental tests. While artificial grass could have been an option, the grass structure in their natural habitat is not uniform. Additionally, due to the small size of these lizards (40 mm SVL), they are difficult to spot in leaf litter and grass, as noted by Cole et al. (1989): '*Spotting the small, shiny brown lizards is a bigger problem*.'

To record the sprint during the 1.0 m, we used a digital camera (Casio Exilim Ex-FH20 9.1MP) attached to a tripod about 1.30 m over the track, and we shot at 400 frames for a second. Individuals from each study area were randomly divided into two groups (n = 9 each), with each group assigned to one of the substrate treatments. We decided not to use the same individual for both substrates to keep the data independent and avoid repeating measurements for the same lizard. Additionally, minimizing the handling of individuals helped reduce stress and prevented fatigue, which could negatively affect their performance. By allowing each individual to run on only one substrate and limiting the number of trials per individual, we ensured that their performance remained consistent and unaffected by exhaustion during the experiment. The experiment was conducted with one lizard at a time, and each lizard ran two times in rapid succession at three different temperatures with a two-hour recovery period between runs. The first run was conducted with body temperature at room temperature (26 – 30°C), the second at -5°C relative to the initial temperature, and the third at +5°C relative to the initial temperature. The selection of temperature conditions followed an adaptation of the methodology described by Diele-Viegas et al. (2018) and the recommendations provided by Taylor et al. (2020). While both studies applied or suggested five or more temperature classes, only three temperature conditions were used in this study. However, strict temperature classes were not predefined, as individual lizards exhibited natural temperature variation. Consequently, the experiment effectively tested more than five temperature points. These variations were essential for constructing the thermal performance curve (TPC). The decision to increase body temperature last was based on the suggestion that higher temperatures pose a greater risk to individuals compared to lower temperatures (Taylor et al., 2020). We classified each of the two runs as either 'good' or 'bad' for further analysis. The 'good' run was defined as the one in which the lizard crossed the track with few stops, while the 'bad' run was characterized by the lizard returning to the starting point or stopping several times. Only the 'good' runs were used for analysis. To reach the required temperature, we placed the lizards inside a styrofoam box with an ice bag and with 60W incandescent light bulbs to decrease and increase the body temperature (Fig. 4). The body temperature was measured through the cloaca using an ultrafine thermocouple connected to an automatic thermometer (Omega® HH806AU accuracy 0.3°C). We did not use the surface temperature here to avoid measuring our own temperature, since we needed to hold the specimen for about 30 seconds to be able to measure. The sprint speed was calculated as the mean of 'good' run of each temperature (three different temperatures per lizard) and disregarding moments that they stopped, using Tracker v5.1.5 (available at www.physlets.org/tracker).

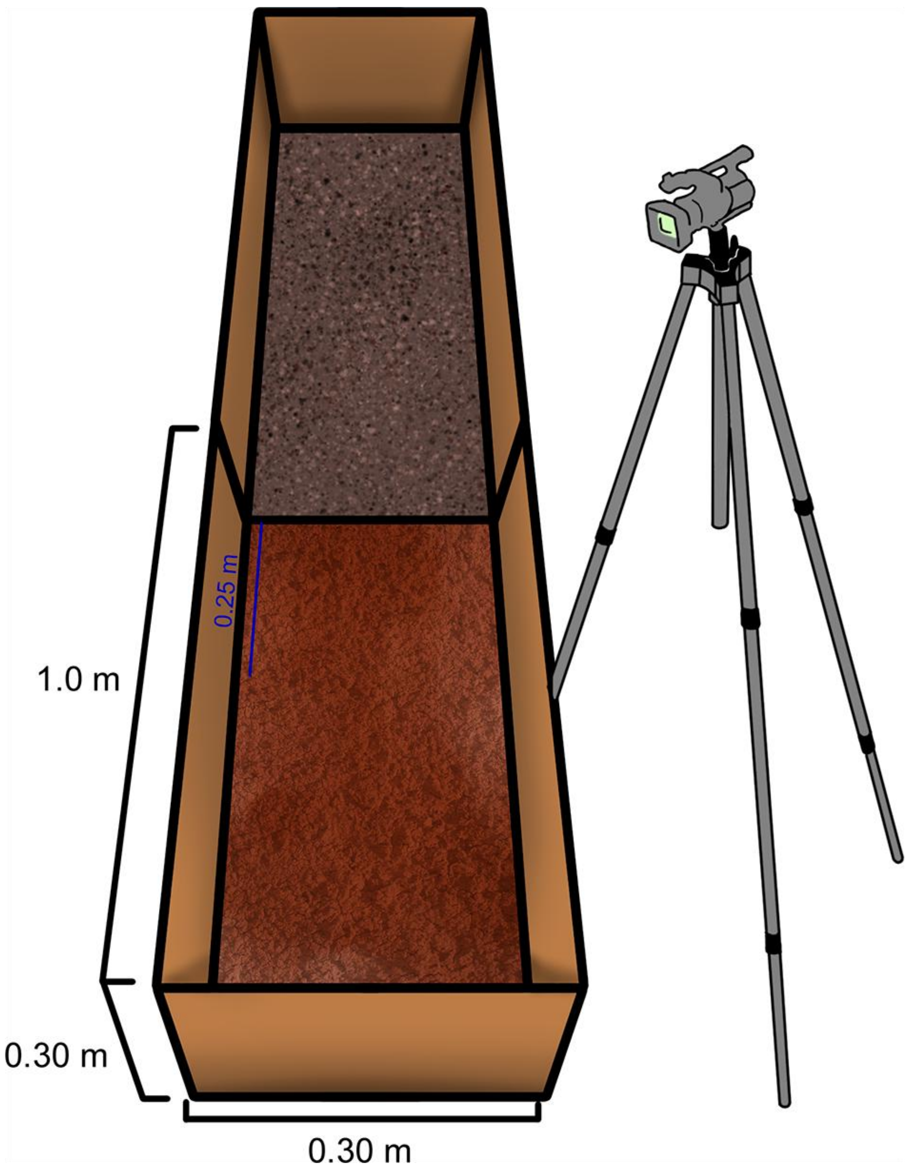

Figure 3. Structure to perform the Sprint performance. Bottom lane treatment with clay (native treatment); Upper lane treatment with sandpaper 60 grit grains ~400 µm (neonative treatment). The track was split in distances of 0.25 m. Drawings by the first author.

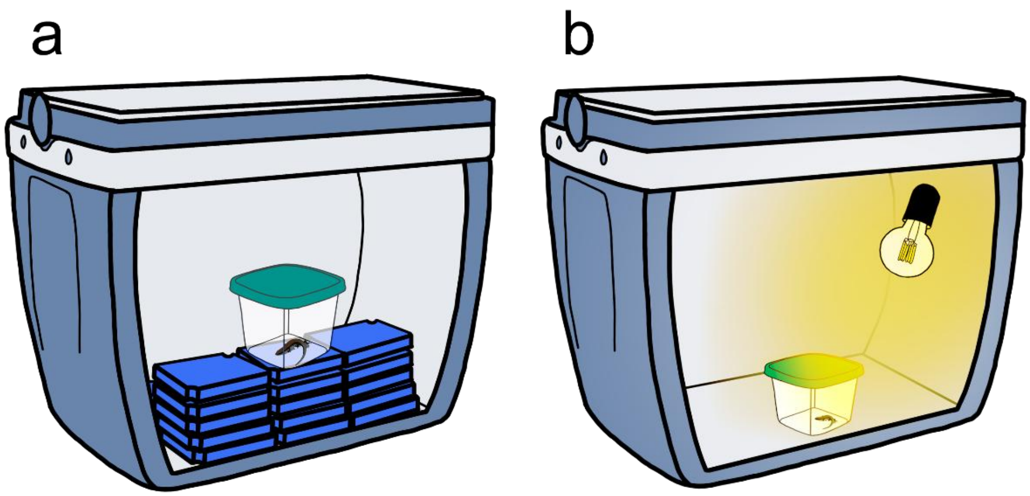

Figure 4. Structure to increase and reduce the body temperature of the individuals. Box with icebags to reduce the body temperature (a); Box with 60w incandescent light bulb to increase the body temperature (b). Drawings by the first author.

1.5. Critical minimum and maximum

Two hours after the final sprint performance test, we proceeded to measure the critical minimum (CT_{min}) and critical maximum (CT_{max}) temperatures, with CT_{max} being the final trait assessed for each lizard. To reach these critical temperatures, we used the same equipment as for sprint performance (described above). CT, was measured when the lizard lost its righting response, while CT_{max} was measured when the lizard exhibited spasms and lost its righting response (Taylor et al., 2020). To reduce the effect of holding the individual, we placed them on a plastic recipient. Between tests, the lizards had a four-hour rest period to recover. Immediately after the temperature measurement, we restore the lizard's body temperature to normal levels using a 60W incandescent light bulb for CT_{min} and a water bath for CT_{max}. The rate of change in temperature was not measured. All individuals were euthanized with xylazine 2%, fixed with 10% formalin, and deposited at the Amphibians and Reptiles Collection from the Instituto Nacional de Pesquisas da Amazônia (INPA-H) at the end of the tests.

1.6. Morphological measurements

To assess the effect of urbanization on morphology and its effect on sprint performance, we measured the morphological traits of individuals. Both specimens collected by our study and specimens previously deposited at the INPA-H Collection (n = 14) were measured. However, only those specimens collected by our study were used to assess the effect of morphology on sprint performance. The specimens were photographed in lateral, dorsal and ventral view, with a digital camera Nikon D3400 with a lens DX 18-55 mm 1:3.5-5.6 G equipped with a generic *close-up* filter (+2) (Figure 5), to measure the traits using ImageJ v1.54b (Schneider et al., 2012; precision of 0.0001 mm). We measured 16 traits (Figure 6) including snout-vent length (SVL); head width (HW); head height (HH); head length (HL); trunk length (TRL); interbrachial-nasal length (INL); femur length (FL); tibia length (TL); hindfoot length (FTL); humeral length (HL); forearm length (FAL); tail length (TAL); pelvic girdle (PG); shoulder girdle (SG); and counted scales of the third finger of hindfoot (SHF), and third finger of forefoot (SFF) (Recoder et al., 2013; Winchell et al., 2018; Vaughn et al., 2021). Furthermore, we measured both the left (_L) and right (_R) sides of each limb to assess fluctuating asymmetry (FA) (Palmer & Strobeck, 1986; Lazić et al., 2013), while for the other analyses we used only the left side. The traits were measured twice on different days and in a different order, and remeasured those that differed by more than 10%. We used just adults in morphological analyses, and we defined them as having

SVL > 35 mm (Hardy et al., 1989) and the development of sexual gonads. The use of ImageJ for measuring animals, particularly lizards, has been validated by Lambert et al. (2012), who demonstrated that its accuracy is comparable to caliper measurements, with a difference of only one standard deviation. Additionally, our measurements differ by 0.7% to 24.3% (Table 2) from those reported by Recoder et al. (2018), who used calipers to measure the same lizard species. While the percentage differences may seem substantial, the absolute values remain closely aligned

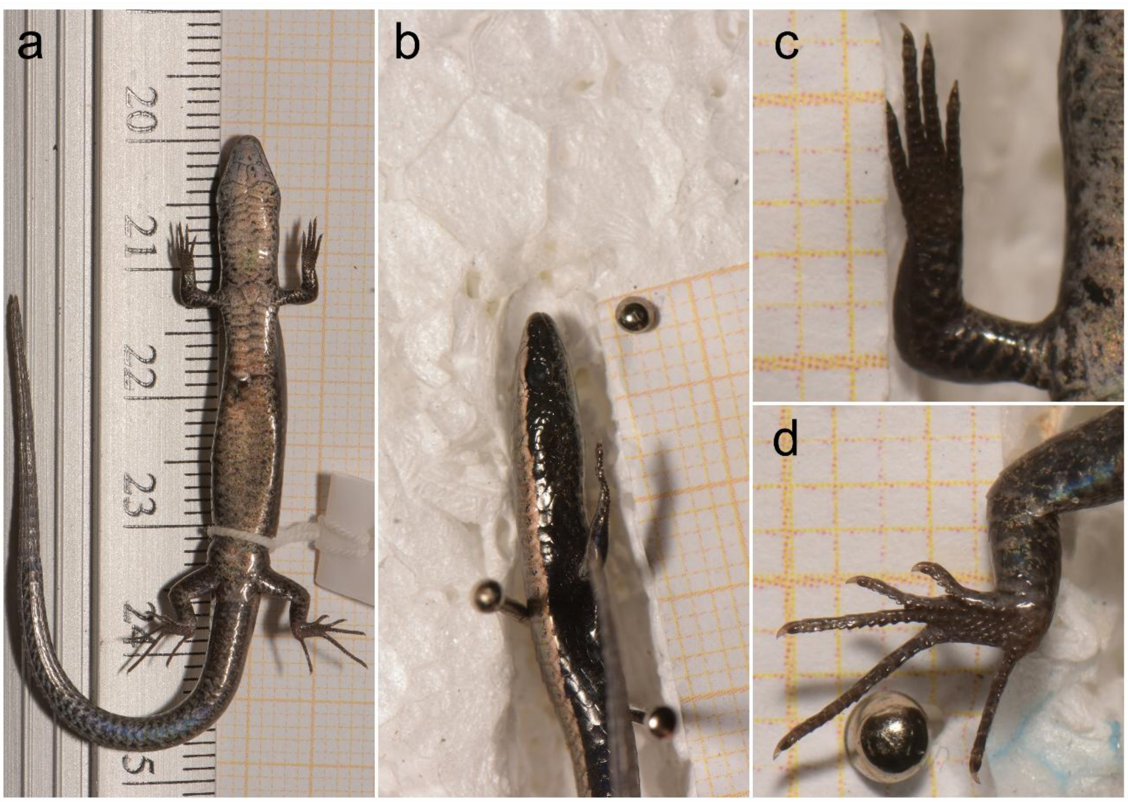

Figure 5. Standardized photographs used for lizard measurements. These images illustrate the photographic methodology used for morphometric and meristic analyses. (a) Ventral view of the

entire body; (b) Lateral view of the entire body; (c) Ventral view with a close-up of the hindfoot for lamellae counting; (d) Ventral view with a close-up of the forefoot for lamellae counting.

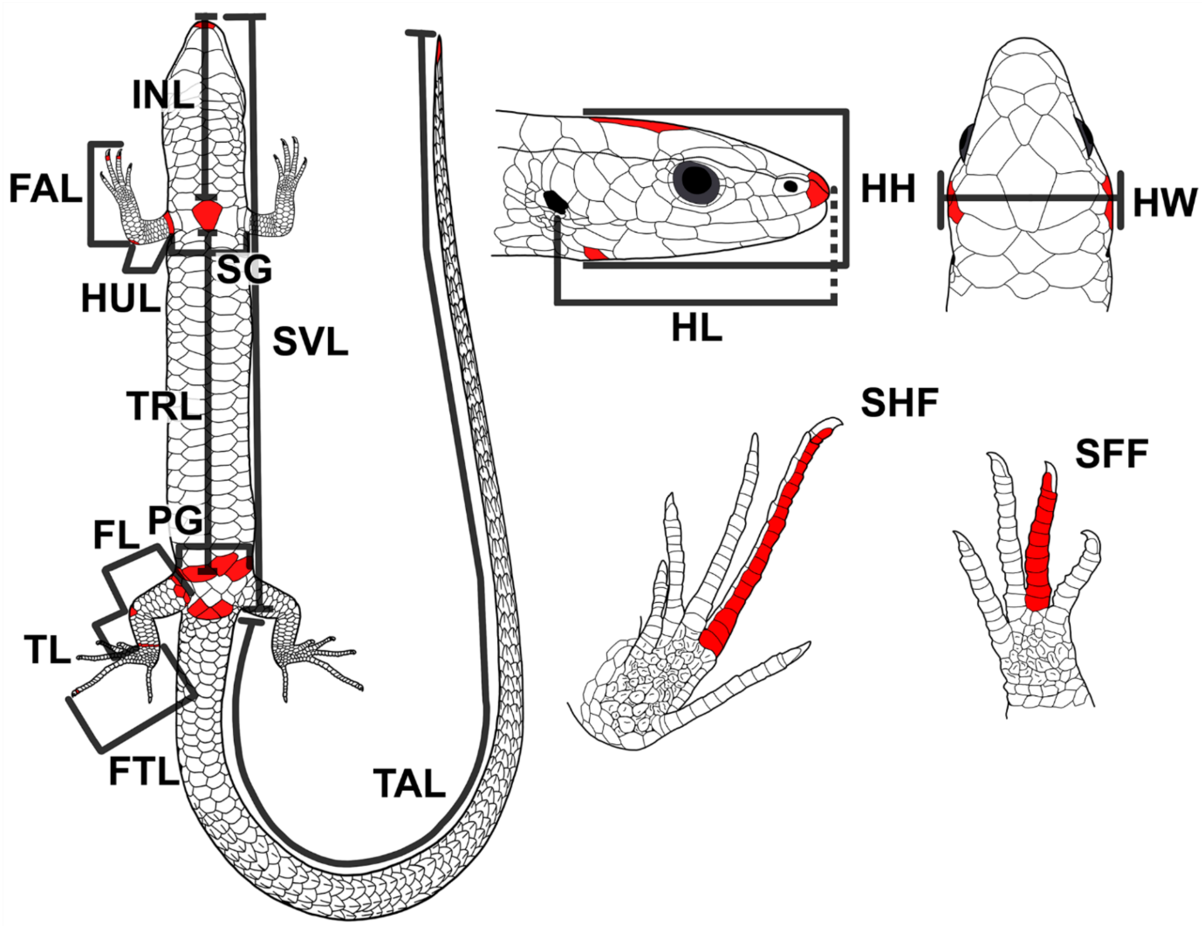

Figure 6. Morphological traits measured. HW: head width; HH: head height; HL: head length; TRL: trunk length; INL: interbrachial-nasal length; HUL: humeral length; FAL: forearm length; FL: femur length; TL: tibia length; FTL: hind foot length; SVL: Snout–vent length; TAL: tail length; PG: pelvic girdle; SG: shoulder girdle; SHF: scales of the third finger of hindfoot; SFF: scales of the third finger of forefoot. Drawings by the first author.

Table 2. A comparison made between the measurements taken by Recoder et al. (2018) using a caliper and those obtained by us using ImageJ v1.54b. HW: head width; HH: head height; HL: head length; TRL: trunk length; INL: interbrachial-nasal length; HUL: humeral length; FAL: forearm length; FL: femur length; TL: tibia length; FTL: hind foot length; SVL: Snout–vent length; TAL: tail length.

Trait	**Recoder et al. 2018 (n=16)**	**This work (both population, juveniles and adults; n = 75)**	**Difference between average values in %**
SVL	36.2 (26.6-41.7) mm	38.7 (25.2-48.6) mm	6.8
TRL	20.9 (15.4-25.3) mm	20.7 (12.6-27.0) mm	1.0
HH	2.9 (2.3-3.6) mm	3.5 (2.6-4.3) mm	21.3
HW	4.5 (3.9-5.0) mm	5.3 (3.7-6.4) mm	17.8
HL	6.6 (5.3-7.3) mm	6.5 (5.1-7.7) mm	1.2
FEM/FL	4.4 (2.7-5.1) mm	4.4 (3.0-5.2) mm	0.7
TIB/TL	3.8 (2.5-4.3) mm	3.1 (1.9-3.9) mm	19.1
FTL	6.2 (5.0-6.7) mm	6.4 (4.4-7.4) mm	3.5
HUM/HUL	2.7 (1.9-3.1) mm	3.1 (2.1-4.1) mm	15.6
FAL	6.3 (4.7-7.3) mm	6.2 (4.5-7.1) mm	1.8
TAL	61.6 (n = 5; 52.5-69.6) mm	46.7 (n = 65; 10.5-75.6) mm	24.3

1.7. Data analysis

All analyses were performed using R software v4.2.1 (R Core Team 2022). The scripts and raw data can be accessed on Dryad (see Main Text). Individuals who died during the tests were not used for thermal analyses, only for morphology.

CT_{min}, CT_{max}, and T_{pref} were compared between the neonative and native populations using function *t-tests*, following Silva et al. (2022) (Table 3). To compare sprint performance between populations, we constructed a thermal performance curve (TPC) using a generalized additive mixed model (GAMM) implemented in the *mgcv* v1.8 package (Wood, 2001). The TPC was generated by setting the velocities at the values of CT_{min} and CT_{max} to zero (Taylor et al., 2020). Morphological traits that exhibited differences between the native and neonative populations were included in the TPC analysis, along with the treatment and geographic distribution, all treated as fixed factors. The curve's smoothness was determined by population as group parameters, and individuals were considered as random factors. Model selection was performed using *MuMin* v1.47.1 package (Bartoń, 2022). We selected the model with an AICc difference smaller than 2 ($\Delta < 2$), from the model with the lowest AICc. Graphs were made using the *ggplot2* package (Wickham, 2011). The temperature at which the lizard reached the highest performance was assumed as the thermal optimum performance (T_{opt}), and the thermal performance breadth was calculated as the range of temperatures where at least 80% of the maximum performance is achieved (B80) (Taylor et al., 2020).

Before comparing the morphology of the neonative and native populations, we assessed the dependence of morphological traits on snout-vent length using a linear model (*lm*). When the relationship was statistically significant, the residuals of the *lm* were used for further analysis. We compared the SVL between neonative and native populations using a t-test. To visualize the morphological differences, we performed a principal component analysis (PCA) using *prcomp* function with the data *z-score* standardized. The results of PCA can be seen in Table S1 and Figure S2. The morphological differences between populations (neonative × native) were assessed using a non-parametric multivariate analysis of variance (NP-MANOVA) with residual randomization in permutation procedure (RRPP; Collyer & Adams, 2018) employing 99999 permutations, following Telemeco and Gangloff (2020). This approach generates empirical sampling distributions by randomly permuting the residuals, reducing the chances of Type-I error and the concern regarding the number of variables and sample size (Telemeco & Gangloff, 2020). The values of the least-square mean result from RRPP can be seen in Table 4.

Table 3. Results of *t.test* of thermal traits between neonative and native populations. The normality of data was assessed using Shapiro-Wilk and LeveneTest. T_{pref}: preferential temperature; CT_{max}: critical maximum temperature; CT_{min}: critical minimum temperature.

						t_test		
Trait	**Population**	**N**	**Mean**	**sd**	**se**	**t**	**df**	**p-value**
T_{pref}	Neonative	18	32.7078	1.8213	0.4293	0.8658	33	0.3928
	Native	17	32.2435	1.29866	0.3150			
CT_{max}	Neonative	18	45.0500	0.8459	0.1994	0.8055	34	0.4261
	Native	18	44.8556	0.5772	0.1361			
CT_{min}	Neonative	18	15.5889	1.2024	0.2834	-4.3398	34	0.0001
	Native	18	17.6500	1.6169	0.3811			

		Shapiro-Wilk		**LeveneTest**		
Trait	**Population**	**W**	**p-value**	**df**	**F value**	**Pr(>F)**
T_{pref}	Neonative	0.97413	0.5661	1	0.3635	0.5507
	Native			33		
CT_{max}	Neonative	0.9461	0.0788	1	0.5071	0.4812
	Native			34		
CT_{min}	Neonative	0.95848	0.1929	1	2.0987	0.1566
	Native			34		

Table 4**.** Least-square of predicted values from NP-MANOVA with residual randomization in permutation procedure. All traits were used as residual values of the linear regression between the trait and SVL in PCA. HW: head width; HH: head height; HL: head length; TRL: trunk length; INL: interbrachial-nasal length; HUL: humeral length; FAL: forearm length; FL: femur length; TL: tibia length; FTL: hindfoot length; TAL: tail length; PG: pelvic girdle; SG: shoulder girdle; SHF: scales of the third finger of hindfoot; SFF: scales of the third finger of forefoot.

	Neonative			**Native**		
Trait	**Mean**	**95% lower confidence**	**95% upper confidence**	**Mean**	**95% lower confidence**	**95% upper confidence**
HL	-0.0186	-0.3750	0.3614	0.0241	-0.3904	0.4743
HH	0.3590	0.0295	0.6917	-0.5032	-0.8936	-0.1097
HW	-0.0750	-0.4407	0.2865	0.1056	-0.3262	0.5349
TRL	0.3886	0.0731	0.7226	-0.5448	-0.9164	-0.1514
INL	-0.4722	-0.7813	-0.1739	0.6616	0.3007	1.0174
SG	-0.1052	-0.4685	0.2607	0.1503	-0.2755	0.5824
PG	0.1100	-0.2419	0.4793	-0.1528	-0.5683	0.2882
FL	-0.2448	-0.5887	0.1085	0.3415	-0.0669	0.7629
TL	0.1897	-0.1635	0.5538	-0.2657	-0.6802	0.1581
FTL	0.5835	0.3189	0.8456	-0.8155	-1.1272	-0.5057
HUL	-0.0569	-0.4430	0.2925	0.0814	-0.3729	0.4895
FAL	0.4449	0.1314	0.7511	-0.6216	-0.9924	-0.2622
SHF	0.3364	-0.0014	0.6586	-0.4706	-0.8736	-0.0768
SFF	0.1112	-0.2530	0.4739	-0.1526	-0.5860	0.2784
TAL	-0.0468	-0.4194	0.3184	0.0647	-0.3685	0.4937

To assess the fluctuating asymmetry (FA), we followed Lazić et al., (2013). First, we calculated an asymmetry index (AI) for all traits as the value on the right side minus on the left side ($AI_{trait1} = R_{side} - L_{side}$) (see mean values in Table S4), where positive values suggest right dominance and negative values suggest left dominance. Second, we assessed trait size dependence on total body size using a linear regression between asymmetry index and SVL (AI ~ SVL) (Table 5) and the dependence on own trait size using a linear regression between asymmetry index and the average between left and right measures of each trait [AI ~ $(R + L)/2$] (Table 6). Third, we assessed whether distributions have directional asymmetry (DA) or FA through a linear mixed-effect model (LMM) with the package *nlme* (Pinheiro et al., 2023). The LMM was performed with the original data for each population for each morphological trait separately. Morphological traits were set as fixed factors between side and their interaction with specimens (side:specimen), both as fixed factors, and specimen measurements as random factors. For the count of scales of hindfoot and forefoot (SHF and SFF) we used the family poisson in *glmmTMB* package (Brooks et al., 2017). Finally, we compared the FA between the native and neonative populations using a one-way ANOVA. For the comparison, we calculate the absolute AI values with averages of each morphological trait value corrected using the log10 transformation ($AI_{trait1} = |\log(\bar{x}\, R_{side}) - \log(\bar{x}\, L_{side})|$)

Table 5. Results of analysis of variance (ANOVA) assessing the relation of asymmetry index (AI) of morphology measures with snout-vent length (SVL). HUL: humeral length; FAL: forearm length; FL: femur length; TL: tibia length; FTL: hindfoot length; SFT: scales of the third finger of hindfoot; SHF: scales of the third finger of forefoot; Sd: standard deviation; Se:

standard error; Df: degrees of freedom.

Trait	Df	Sum sq	Mean sq	F value	Pr(>F)
FL	1	0.1862	0.1862	2.3018	0.1323
TL	1	0.0077	0.0077	0.1510	0.6984
FTL	1	0.2090	0.2090	2.2864	0.1336
HUL	1	0.0234	0.0234	0.5389	0.4646
FAL	1	0.0997	0.0997	1.3193	0.2534
SHF	1	0.1950	0.1947	0.5318	0.4675
SFF	1	0.1500	0.1500	0.3687	0.5450

Table 6. Results of analysis of variance (ANOVA) assessing the relation of asymmetry index (AI) of morphology measures with trait size $(Right_{side} + Left_{side})/2$. HUL: humeral length; FAL: forearm length; FL: femur length; TL: tibia length; FTL: hind foot length; SFT: scales of the third finger of hindfoot; SHF: scales of the third finger of forefoot; Sd: standard deviation; Se: standard error; Df: degrees of freedom.

Trait	Df	Sum sq	mean sq	F value	Pr(>F)
FL	1	0.2241	0.2241	2.7839	0.0983
TL	1	0.1140	0.1140	2.2850	0.1337
FTL	1	0.2155	0.2155	2.3594	0.1276
HUL	1	0.1609	0.1609	3.8294	0.0531
FAL	1	0.2085	0.2085	2.7984	0.0974
SHF	1	0.0340	0.0345	0.0937	0.7601
SFF	1	0.0410	0.0411	0.1007	0.7516

1.8. Inferring species distribution and extinction risks

To assess the current distribution of *Gymnophthalmus underwoodi*, infer its potential dispersion routes and its local extinction risks, we performed a hybrid species distribution model. We chose the hybrid model since this approach has been shown as a more complete method to assess the effects of climate change on species (Meineri et al., 2015; Caetano et al., 2020; Tourinho & Vale, 2023).

We selected four types of predictors: (1) Bioclimatic variables, available at WorldClim (https://www.worldclim.org/) (Fick & Hijmans 2017); (2) Soil content (percentage of clay, silt, and sand) available in Soilgrid (https://soilgrids.org/) (Hengl et al., 2017) for 0.0 and 5 cm depths; (3) Landcover based on Global land-use and land cover change (LUCC) (available at https://geosimulation.cn/GlobalLUCCProduct.html) (Li et al., 2017); (4) Slope acquired through altitude using function *terrain*; and (5) Ecophysiological variables, the hour of activity (H_a) and sprint performance. We excluded four bioclimatic variables that showed spatial anomalies in the study area (Bio07, Bio16, Bio18, and Bio19). For the remaining 15 variables left, we used variance inflation factor (VIF) with a threshold of 10 to reduce multicollinearity while changing Bio02 (Mean Diurnal Range) for Bio05 (Maximum Temperature of Warmest Month), and Bio12 (Annual Precipitation) for Bio15 (Precipitation Seasonality) since extreme are important to assess species vulnerability (Román-Palacios & Wiens, 2020). The landcover was used as a continuous raster, where all forest biomes were set as 1 and grassland as 0. We used two scenarios of forest predictions in the future models, where the pessimistic scenario A2 was used for SSP5-8.5 and moderate scenario A1B for SSP3-7.0 (Sales et al., 2020). The ecophysiological variables were derived from the functional data we collected and geographically projected using the microclim method in the Mapinguari package (https://gabrielhoc.github.io/Mapinguari.html;

Caetano et al., 2020) and the microclim database of Kearney et al. (2014). We used the TPC calculated before (see Data Analysis) since this is a proxy to Darwinian fitness usually used for lizards (Angilletta 2009; Taylor et al., 2020), and hours of activity (H_a), an important predictor for assessing the effects of climate change (Caetano et al., 2020; Sinervo et al., 2010). All variables were aggregated to 2.5' (~ 5 km) spatial resolution. The correlation among all predictors was assessed using VIF (>10), resulting in the final 11 variables selected (Dormann et al., 2013; James et al., 2021; Sillero et al., 2021). We predicted the species distribution under two climate change scenarios: moderate (SSP3-7.0) and extreme (SSP5-8.5), for the years 2060 and 2100 (Riahi et al., 2017; IPCC 2021). To account for uncertainty, we choose four different global climate models (IPSL-CM6A-LR, MIROC6, MRI-ESM2-0, and UKESM10LL) that represent a good projection in the study area (Cannon, 2020).

We delimited the accessible area with a minimum convex polygon with a 500 km buffer (Barve et al., 2011; Mendes et al., 2020; Sillero et al., 2021). This area almost covers the entire distribution of the genus *Gymnophthalmus,* thus covering a large variability of climate conditions available (Elith et al., 2010; Escobar et al., 2014; Sillero et al., 2021). However, we extrapolate the model to an area that ranges from latitude -90º to -45º, and longitude -10º to 25º to evaluate possible future areas of dispersion. The distribution records used were obtained through our own field records, literature records, GBIF (available at https://www.gbif.org/), and iNaturalist (available at https://www.inaturalist.org/), using only unique occurrences (157 total) (available on Dryad). We used an environmental filter, *EnvSample* function, that reduces environmental bias using the grid resolution of the variables (Varela et al., 2014), only with predictors selected with VIF<2 (seven in total) from those 11 sets above, leaving 103 records. Additionally, we used a 10 km buffer (Castellanos et al., 2019) to reduce geographic biases, keeping 75 records.

We performed the SDM with package *ENMTML* (Andrade et al., 2020) using four algorithms: Maximum Entropy (presence and background) (Phillips et al., 2006), Bayesian Gaussian Process (presence and pseudo-absence) (Golding, 2014; Golding & Purse, 2016), Random Forest (presence and pseudo-absence) (Liaw & Wiener, 2002), and Support Vector Machine (presence and pseudo-absence) (Karatzoglou et al., 2004). The final model was generated by the ensemble of models by the weighted mean of Sørensen index (Leroy et al., 2018), maximizing the true positives to evaluate the models. This increases the capacity of discriminating between presence and absence sites, avoiding bias by the prevalence of data (ratio between species occupancy area and study area) (Leroy et al., 2018). The choice of the four algorithms was based on their general performance in default settings and in the type of data necessary for their execution (absence and background), allowing a better ensemble (Sillero et al., 2021; Valavi et al., 2022; 2023). In addition, we performed the model using the Bioclim and generalized linear models algorithms (also available in *ENMTML*), however, they did not perform well (low values of AUC, TSS, and Sørensen), so we excluded them from the analysis. The number of pseudo-absences represents fake locations where the species is not observed, being equal to presences (ratio equal to 1) (Andrade et al., 2020). These pseudo-absences were allocated outside a 50 km buffer, following the default parameters of the package (Barbet-Massin et al., 2012). The background points are the locations for climate sampling of the environment, and the number was determined following the default parameter (10,000) (Andrade et al., 2020). The data partition was made using the 'checkerboard' cross-validation approach to reduce the spatial correlation (Roberts et al., 2017). Sørensen and TSS metrics (Allouche et al., 2006; Leroy et al., 2018), were used for model evaluation in which values close to +1 suggest a good model.

The model ensemble resulted in a threshold of 0.32 (Table 7) that was used to make a binary plot and estimate the area of occurrence. The estimated area of occurrence was calculated using the same polygon from the accessible area. The current and future binary plots were overlapped to assess which areas are considered stable, and in which areas *Gymnophthalmus underwoodi* is forecast to lose or gain ranges.

Table 7. Evaluation table of Species Distribution Model for *Gymnophthalmus underwoodi*. The ensemble model was made using the weighted mean of the Sørensen index. Threshold was calculated using the Sørensen index. AUC was not used to select the best model however we chose to inform since it is a common evaluation metric. MXD: Maximum Entropy default (all features); SVM: Support Vector Machine; RDF: Random Forest; GAU: Bayesian Gaussian Process.

Algorithm	**Threshold**	**AUC**	**TSS**	**Sørensen**
MXD	0.99	0.82	0.60	0.78
SVM	0.71	0.86	0.66	0.84
RDF	0.59	0.83	0.58	0.80
GAU	0.60	0.84	0.62	0.81
Ensemble	0.32	0.86	0.70	0.84

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